

Arnav

📧 [REDACTED] | @ [REDACTED] | 🌐 Cupertino, CA

SKILLS

Cloud & Deployment: AWS (EC2, Lambda, S3, DynamoDB), Azure, Firebase, Docker, Linux

Languages: Java, Python, C, C++, C#, TypeScript/JavaScript, SQL

Backend & Web: PostgreSQL, Redis, Node.js, .NET, React, HTMX, Tailwind CSS

ML & Computing: CUDA, OpenMP, TensorFlow, PyTorch, OpenCV

WORK EXPERIENCE

[REDACTED]
Backend Engineer / Java, Python, Linux

San Jose, California

November 2025 – Present

- Developed a web crawler to collect healthcare provider data from major news publishers and improve provider profiles.
- Implemented a point-based matching system to link provider records to government datasets, improving data completeness and attribution.
- Created a validation and cleaning pipeline to standardize provider attributes and reduce incorrect records.
- Designed a regression test harness to check scraping and parsing logic.
- Tuned attribute matching weights with Bayesian optimization, improving provider extraction accuracy by 8% across multiple datasets.
- Standardized CLI guidelines and argument parsing to improve team workflows.

Odoo

Full Stack Developer Intern / Python, JavaScript, PostgreSQL

Buffalo, New York

June 2023 – September 2023

- Implemented a 4-4-5 accounting schedule module so businesses could manage custom fiscal periods and improve reporting accuracy.
- Optimized SQL queries and database access to improve fiscal schedule data retrieval.
- Wrote unit tests with 90% coverage for default modules used in client solutions and refactored code for easier maintenance.

PROJECTS

Virtual Cycling Leaderboard | *Azure, Firebase, Docker, .NET, Python*

- Connected riders in virtual cycling rooms through BLE sensors to share speed, distance, heart rate, and cadence in real time.
- Deployed a .NET API service that converts rider stats into downloadable FIT files for sports tracking apps.

Deep Neural Network Library | *C++, CUDA, OpenMP*

- Designed a C++ neural network library with a PyTorch-like API and multiple activation and cost functions for ML experiments.
- Wrote CUDA kernels for matrix operations and compared serial, parallel CPU, and GPU training, achieving a 6x speedup.

Command Finder with MCP | *Python, unittest, MCP*

- Developed a Python CLI and Model Context Protocol (MCP) server to store and retrieve common deployment commands.

EDUCATION

Santa Clara University

M.S. in Computer Science and Engineering

Santa Clara, CA

September 2024 – June 2025

Santa Clara University

B.S. in Computer Science and Engineering

Santa Clara, CA

September 2020 – June 2024

- **Activities:** President of Triathlon Club
- **Awards:** 2022 Santa Clara Inrix Hackathon Winner, 2024 Collegiate Triathlon All-American